

Model Performance Report for Stock Price Prediction

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Abstract

This report presents the process and results of predicting Apple (AAPL) stock closing prices using traditional and advanced machine learning models. Three benchmarks were conducted, focusing on data preprocessing, model development, and hyperparameter tuning. The Linear Regression, Decision Tree, Random Forest, LSTM, GRU, and CNN models were evaluated to determine the best approach. After extensive testing and tuning, the LSTM model emerged as the most accurate, providing the lowest Mean Squared Error (MSE) compared to other models.

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Benchmark 1: Data Preprocessing and Exploratory Data Analysis

The goal of Benchmark 1 was to prepare the data for modeling. This meant cleaning the data, handling missing values, and creating visualizations. Below are the steps I took:

1. Data Collection

Data was collected for Apple (AAPL) from January 1, 2020, to January 1, 2024, using the yfinance library. The dataset includes daily prices and other relevant information about Apple stock.

2. Data Cleaning

After downloading the data, missing values were checked using the `.isnull().sum()` function. Missing values can create issues in modeling, so identifying and handling them properly is important. Fortunately, there were no significant missing values, indicating that the data was mostly clean. This can be seen in Figure 1.

3. Data Visualization

To better understand the data, the following visualizations were created:

- **Figure 2: Closing Prices:** A plot was created showing Apple's closing prices from 2020 to 2024. This visualization helped identify changes in the stock price over time.
- **Figure 3: Moving Averages:** The 5-day and 20-day moving averages for closing prices were calculated and plotted. These moving averages helped highlight trends by smoothing out short-term price fluctuations.

4. Summary Statistics

Summary statistics, including average, minimum, and maximum values for columns such as closing price, opening price, and volume, were generated. These statistics provided an understanding of the range and distribution of the data.

5. Insights

The visualizations and statistics provided the following insights:

- There were periods of rapid growth as well as declines in the stock price.
- Moving averages highlighted trends in Apple's stock performance more clearly.

Overall, Benchmark 1 focused on understanding the data and preparing it for modeling. With the data cleaned and visualized, it is now ready for the next steps, which involve developing machine learning models to predict stock prices.

```
PS C:\Users\krist\OneDrive\Desktop\EECS 658\Honors Contract Assignment> & C:/Python312/python.exe "
[*****100%*****] 1 of 1 completed
Raw Data:
Price          Date Adj Close      Close      High      Low      Open      Volume
Ticker
0      2020-01-02 00:00:00+00:00 72.876114 75.087502 75.150002 73.797501 74.059998 135480400
1      2020-01-03 00:00:00+00:00 72.167603 74.357498 75.144997 74.125000 74.287498 146322800
2      2020-01-06 00:00:00+00:00 72.742668 74.949997 74.989998 73.187500 73.447502 118387200
3      2020-01-07 00:00:00+00:00 72.400558 74.597504 75.224998 74.370003 74.959999 108872000
4      2020-01-08 00:00:00+00:00 73.565208 75.797501 76.110001 74.290001 74.290001 132079200

Missing Values:
Price      Ticker
Date
Adj Close  AAPL      0
Close      AAPL      0
High       AAPL      0
Low        AAPL      0
Open       AAPL      0
Volume     AAPL      0
dtype: int64
```

Figure 1: Raw Data and Data Cleaning



Figure 2: Closing Prices



Figure 3: Moving Averages

Benchmark 2: Developing and Evaluating Machine Learning Models

The goal of Benchmark 2 was to develop and evaluate traditional machine learning models. Three models were used: Linear Regression, Decision Trees, and Random Forests. The models predicted the next day's closing price of Apple stock (AAPL).

1. Data Preparation

The dataset used for this benchmark included Apple stock prices from January 1, 2020, to January 1, 2024. The feature used was the current closing price, and the target was the next day's closing price. The data was split into training (80%) and testing (20%) sets.

2. Model Development

- **Figure 4: Linear Regression:** A simple model that finds the relationship between the current price and the next day's price.
- **Figure 5: Decision Tree:** A model that splits the data into branches based on decision rules to predict the next price.
- **Figure 6: Random Forest:** A model that uses multiple decision trees to improve accuracy and reduce overfitting.

3. Model Evaluation

The models were evaluated using two metrics:

- **Mean Squared Error (MSE):** Measures how much the predicted values differ from the actual values. Lower is better.
- **R-squared (R^2):** Shows how well the model explains the variation in the data. Values closer to 1 are better.

4. Results

- **Linear Regression:** $MSE = 6.97$, $R^2 = 0.99$
- **Decision Tree:** $MSE = 15.81$, $R^2 = 0.99$
- **Random Forest:** $MSE = 12.61$, $R^2 = 0.99$

The Linear Regression model had the lowest MSE, meaning it made the most accurate predictions. All three models had a high R^2 value, which means they all explained the data well.

5. Insights

- Linear Regression performed best with the lowest error.
- Random Forest and Decision Tree also gave good results but had higher errors compared to Linear Regression.

Benchmark 2 focused on using different machine learning models to predict stock prices. Linear Regression was the most accurate in this case. These models will be compared to more advanced methods in the next benchmarks.

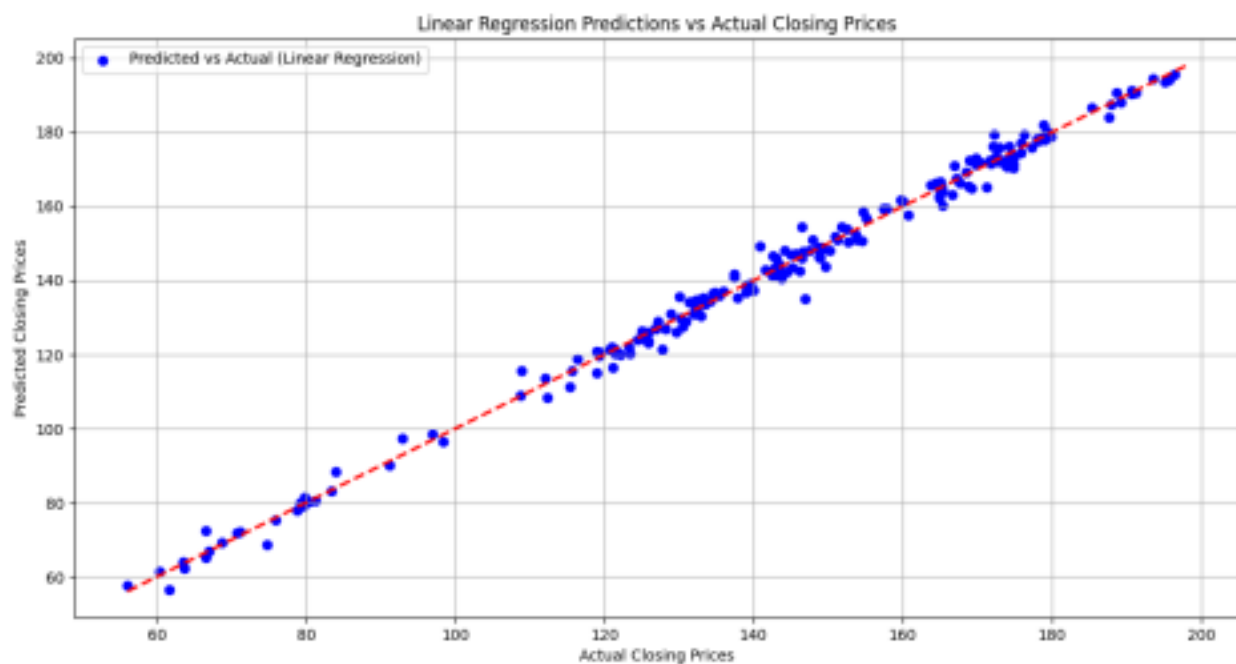


Figure 4: Linear Regression

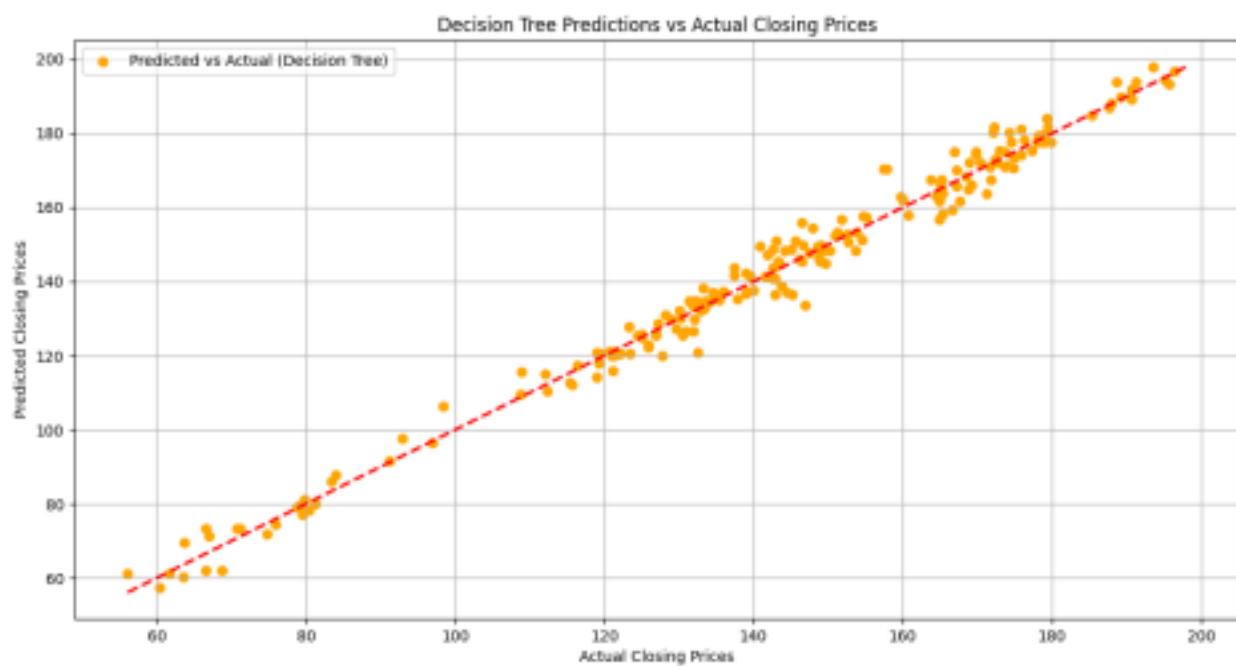


Figure 5: Decision Tree



Figure 6: Random Forest

Benchmark 3: Implementing and Testing Advanced Models Before Hyperparameter Tuning

The goal of Benchmark 3 was to use advanced models like LSTM, GRU, and CNN to predict stock prices. I used time series forecasting methods to predict Apple stock (AAPL) closing prices.

1. Data Preparation

As I did through this project, I downloaded Apple stock data from January 1, 2020, to January 1, 2024. The closing prices were normalized to make the data suitable for training. For this benchmark, I used a sequence length of 10 days to predict the next day's closing price. The data was split using 5-fold cross-validation to evaluate model performance.

2. Models Developed

Three models were used:

- **LSTM (Long Short-Term Memory):** A type of recurrent neural network that works well with time series data.
- **GRU (Gated Recurrent Unit):** Similar to LSTM, but with fewer parameters, making it more efficient.

- **CNN (Convolutional Neural Network):** A one-dimensional CNN model used to identify patterns in the data.

3. Model Evaluation

The models were evaluated using cross-validation. The performance metrics were:

- **Mean Squared Error (MSE):** Measures prediction error and lower values are better.
- **R-squared (R^2):** Shows how well the model explains the variation in the data. Values closer to 1 are better.

4. Results Before Hyperparameter Tuning

- **LSTM:** Average MSE = 0.0028, Average R^2 = 0.9382
- **GRU:** Average MSE = 0.0021, Average R^2 = 0.9542
- **CNN:** Average MSE = 0.0026, Average R^2 = 0.9431

The GRU model performed the best overall, with the lowest error and highest R^2 value. The CNN and LSTM models also gave good results, but the GRU was slightly better.

5. Insights

- GRU provided the best balance between accuracy and efficiency.
- All three models had high R^2 values, meaning they captured most of the trends in the data well.
- During the evaluation, figures were generated for each model comparing predicted prices to actual prices for both the first and last folds.
 - For the **LSTM model**, Figure 7 showed that the predictions had some deviations from the actual values, indicating lower accuracy. However, by Fold 5 in Figure 8, the predictions were much closer to the actual values, showing an improvement in performance.
 - For the **GRU model**, Fold 1 in Figure 9 already showed relatively accurate predictions in comparison to Figure 7, but by Fold 5 in Figure 10, the predicted line almost completely overlapped with the actual price line. This shows to be the strongest performance among all models and folds.
 - For the **CNN model**, the predictions in Fold 1 in Figure 11 were less accurate than Figure 9 and very similar to Figure 7, with larger deviations from the actual values. In Fold 5 in Figure 12, the predictions improved significantly and better than using LSTM, but the GRU still demonstrated the most consistent and accurate results overall.

This part of Benchmark 3 focused on testing advanced models before tuning. In the next part, I will perform hyperparameter tuning to further improve the model performances.

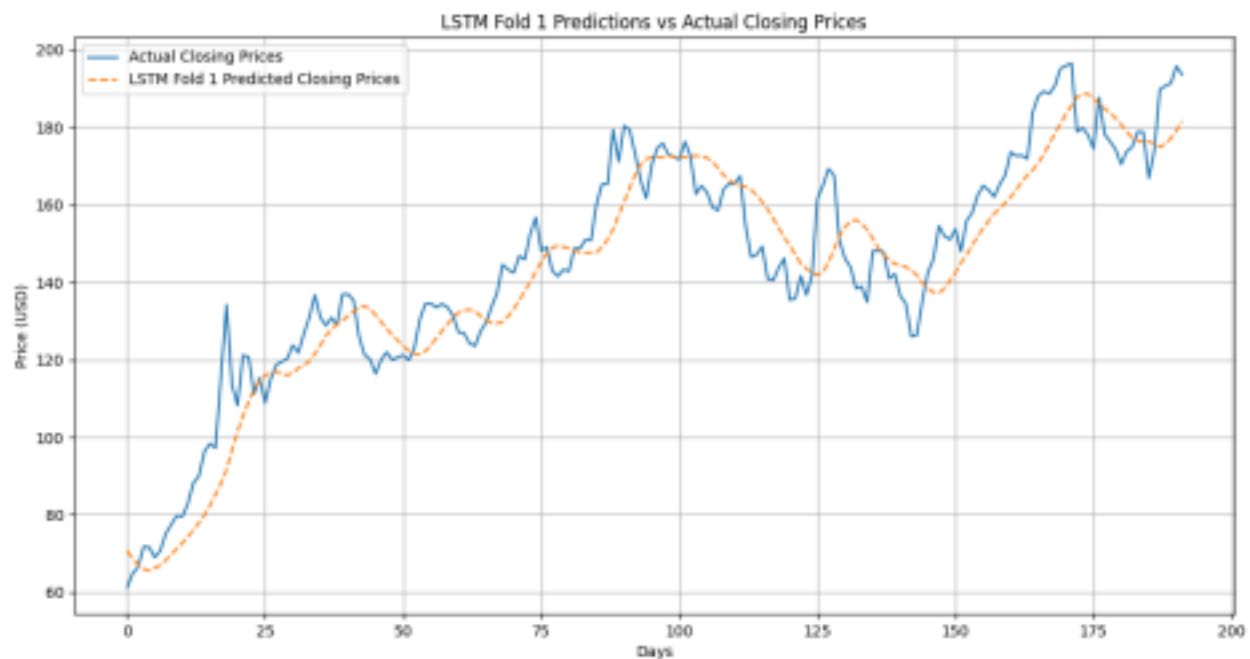


Figure 7: LSTM Fold 1 Graph

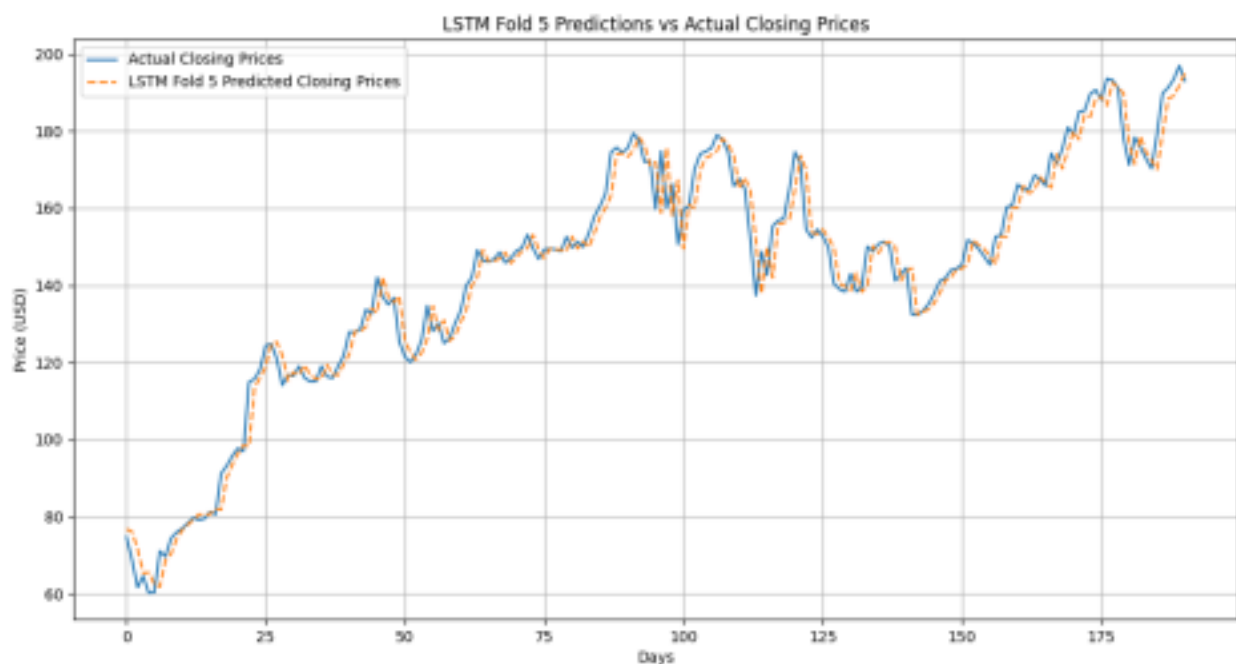


Figure 8: LSTM Fold 5 Graph

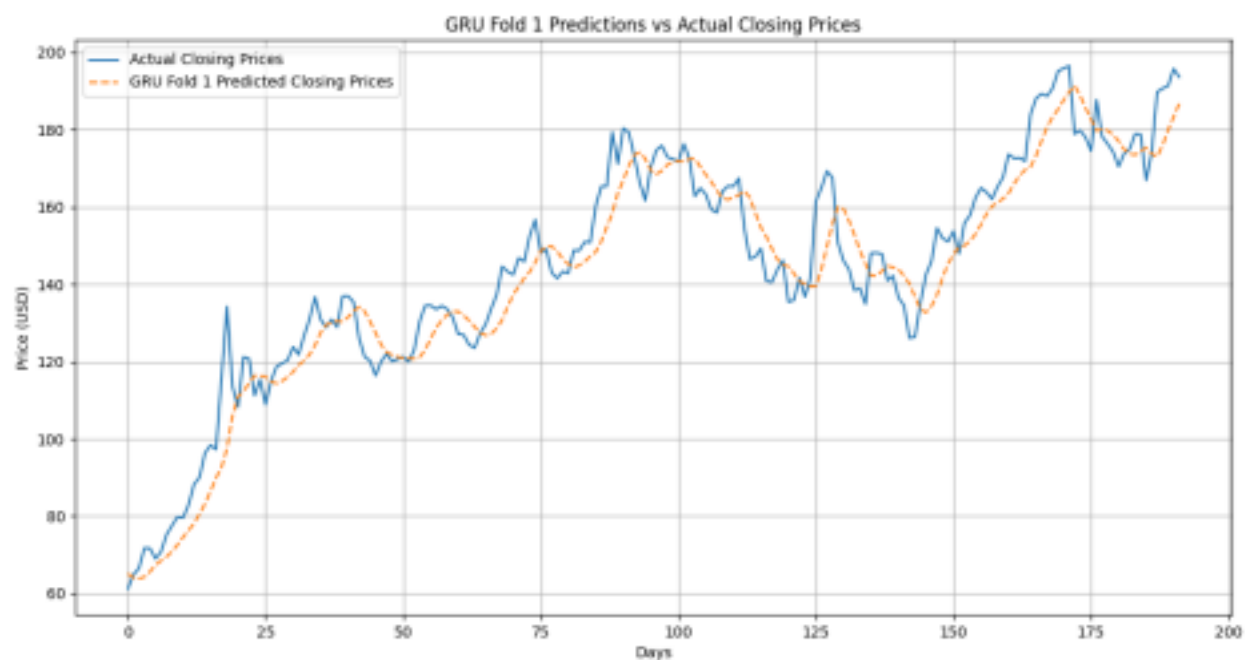


Figure 9: GRU Fold 1 Graph

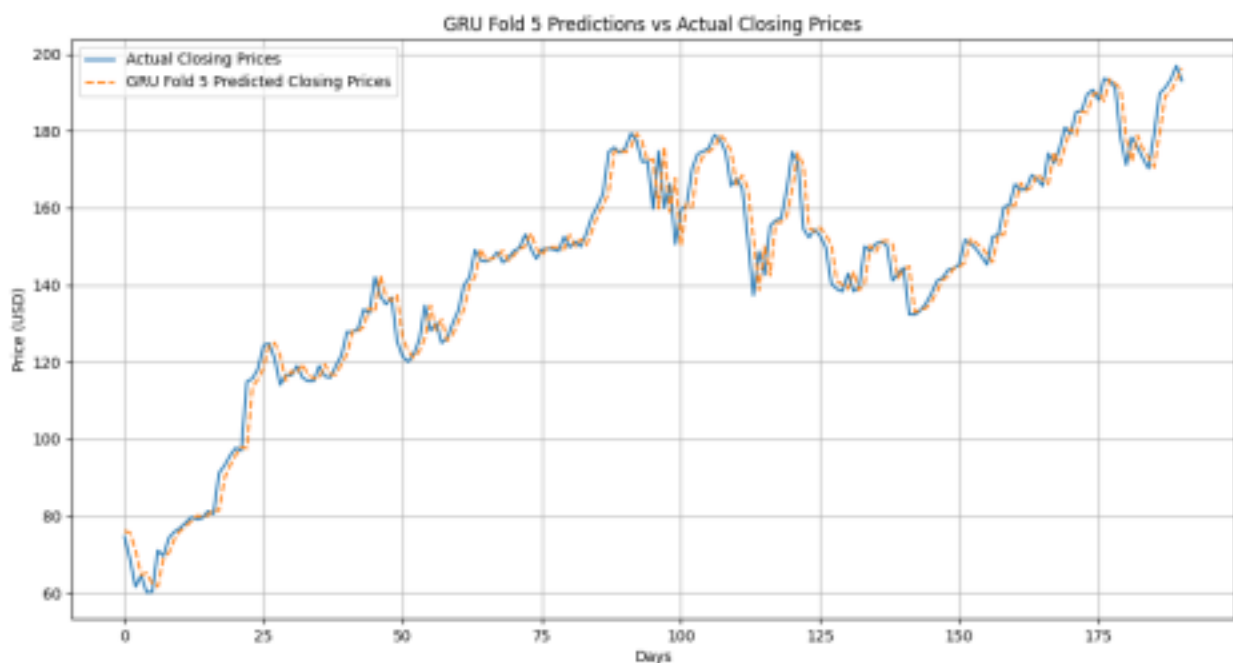


Figure 10: GRU Fold 5 Graph

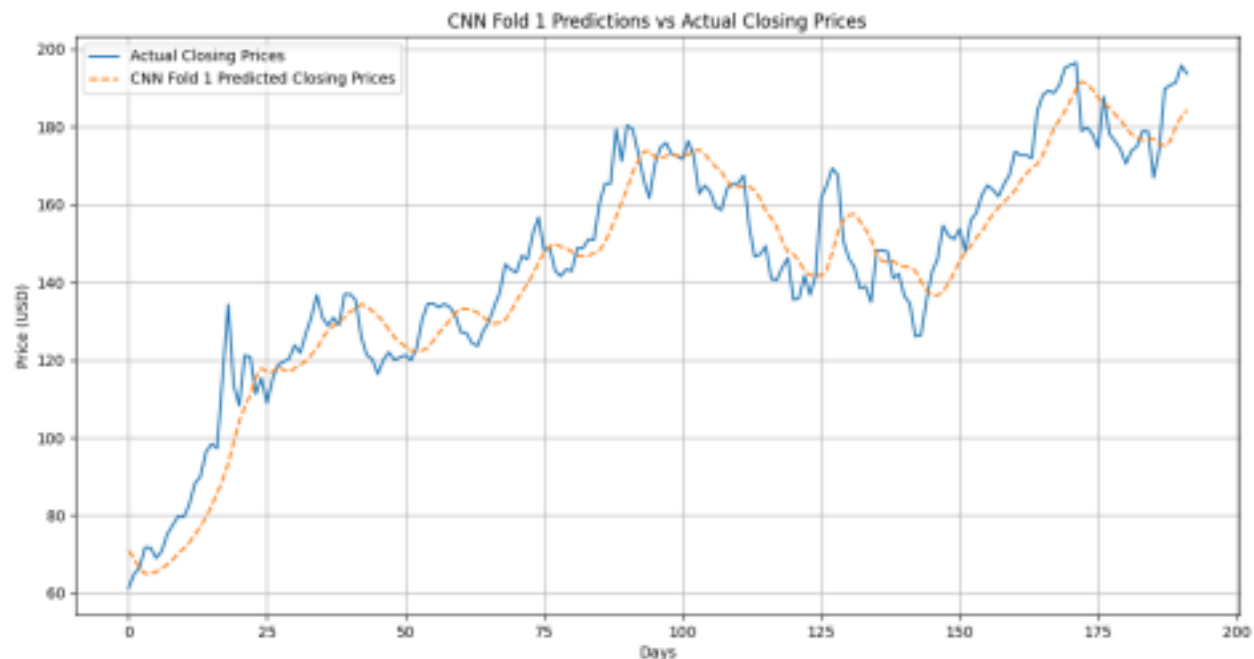


Figure 11: CNN Fold 1 Graph

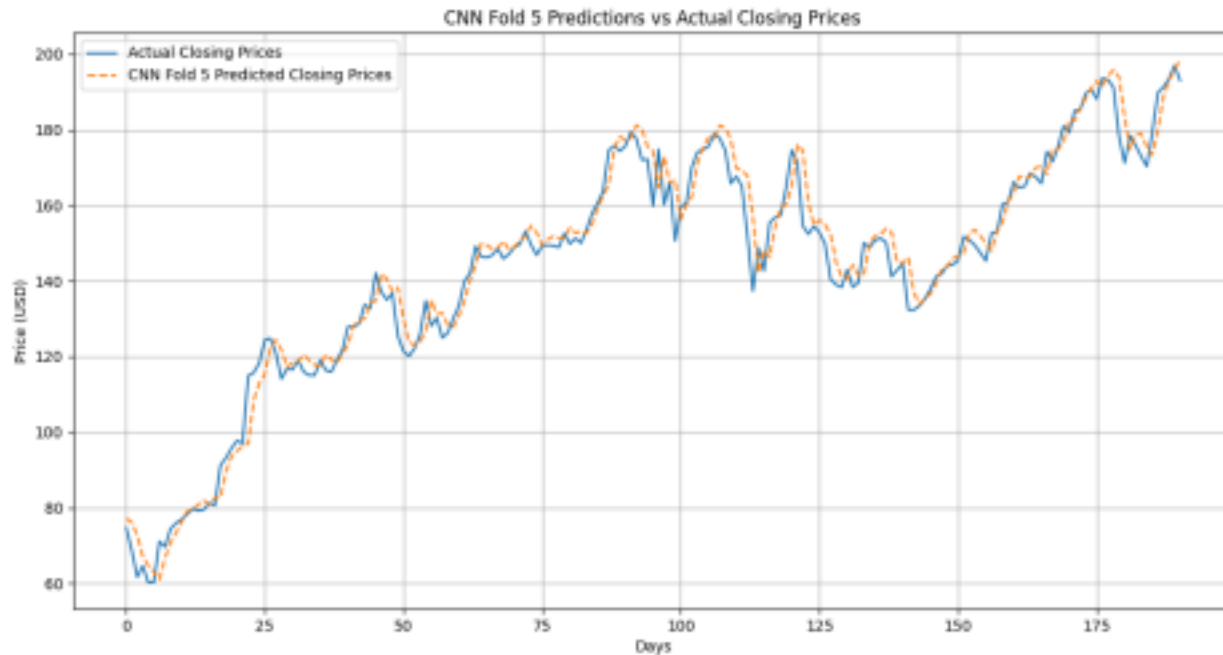


Figure 12: CNN Fold 5 Graph

Benchmark 3: Implementing and Testing Advanced Models After Hyperparameter Tuning

The goal of this section was to improve the LSTM, GRU, and CNN models through hyperparameter tuning and cross-validation. I tested several hyperparameters to find the best combination for each model. The focus was to predict the closing price of Apple stock (AAPL).

1. Hyperparameters Tested

The hyperparameters I tested were:

- **Learning Rate (lr):** Values of 0.0001, 0.001, and 0.01.
- **Hidden Size:** Sizes of 32, 64, and 128 for LSTM and GRU.
- **Number of Layers:** 1, 2, and 3 layers for LSTM and GRU.
- **Batch Size:** 16, 32, and 64.

I used 5-fold cross-validation to evaluate each combination and determine the best settings.

2. Results and Best Parameters

After testing all combinations, the **best model** was an **LSTM** with the following parameters:

- **Learning Rate:** 0.01
- **Hidden Size:** 128
- **Number of Layers:** 3
- **Batch Size:** 16

This model had the lowest Mean Squared Error (MSE) of **0.0023** during cross-validation, indicating the most accurate predictions among all models and parameter combinations. The results of this can be found in **benchmark4.txt**.

3. Model Comparison (Benchmark 2 and 3)

- **Benchmark 2 Results:**
 - **Linear Regression:** Average MSE = 6.97
 - **Decision Tree:** Average MSE = 15.81
 - **Random Forest:** Average MSE = 12.61
- **Benchmark 3 Results Before Tuning:**
 - **LSTM:** Average MSE = 0.0028
 - **GRU:** Average MSE = 0.0021
 - **CNN:** Average MSE = 0.0026
- **Benchmark 3 Results After Tuning:**
 - **Best LSTM Model:** Average MSE = 0.0023

The tuned LSTM model outperformed all other models, both traditional and advanced. It had the

lowest error and was able to capture the trends in the data more accurately.

4. Insights and Challenges

- **Insights:** The best performing model was an LSTM with a higher number of layers and a larger hidden size. This shows that a more complex architecture is beneficial in this case for capturing patterns in stock prices.
- **Challenges:** Tuning hyperparameters required hours to process. Testing different combinations with cross-validation was computationally expensive, but it was necessary to find the optimal settings.

5. Final Conclusion

The **best model** for predicting Apple stock closing prices is the **LSTM** with the tuned hyperparameters (learning rate = 0.01, hidden size = 128, number of layers = 3, batch size = 16). This model provides the most accurate results and has the lowest MSE compared to all other models tested.